Name: _____

WORKSHEET

Counterexamples

1 Provide a counterexample for each statement below to demonstrate it is false.

a All insects have wings.

b February is always the hottest month.

c All pentagons have five obtuse angles.

d All sine graphs start at the origin.

e If $x^2 + 4 = 40$, then $x = 6$.

f $8 - 2x - x^2 \leq 0$ for $x \in [-4, 2]$

2 Decide whether or not the statements below are true or false, providing counterexample for the false ones.

a The minimum for any parabola is found at the origin.

b If a quadrilateral has three right angles, it must be a rectangle.

c A hexagon with sides all equal to 4 cm must be regular and therefore have all its angles equal in size.

d All exponential functions have an unlimited domain.

e $x^2 \geq 0$ for all $x \in \mathbb{R}$.

f If $a^2 - b^2 = 0$, then $a = b$.

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3 In each case below, determine whether or not the counterexample shows that the statement is false.

a Statement: All fish lay eggs.

counterexample: Many sharks give birth to live young.

The statement is not true.

b Statement: All blondes have blue eyes.

counterexample: Ava has blue eyes and brown hair.

The statement is not true.

c Statement: All parallelograms have two pairs of sides equal in length.

counterexample: A kite has two pairs of sides equal in length but is not a parallelogram.

The statement is not true.

d Statement: All similar figures have corresponding sides equal in length.

counterexample: A square and a rhombus have sides equal in length but are not similar.

The statement is not true.

e Statement: $x^2 + 5x - 6 < 0$ for $x \in (-1, 6)$

counterexample: For $x = 6$, $6^2 + 5(6) - 6 = 0$

The statement is not true.

f Statement: $x^2 \leq x^3$ for all $x > 0$

counterexample: For $x = -1$, $\text{LHS} = (-1)^2 = 1$

$\text{RHS} = (-1)^2 = -1$

$1 \not\leq -1$

The statement is not true.

4 For each counterexample in question **3** that does not show that the statement is not true, provide a valid counterexample (where possible).

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Answers

1 **a** Most ants do not have wings.

b In the northern hemisphere, February would never be the hottest month.

c Pentagons exist with one reflex angle and two acute angles and two right or obtuse angles.

d $y = \sin(x + 3)$ does not begin at the origin.

e Another solution is $x = -6$

f For $x = 0$, $8 - 2(0) - 0^2 = 8$.

2 **a** False; $y = x^2 - 10$ has a minimum at $(0, -10)$

b True

c False; a hexagon with six equal sides may contain a reflex angle and therefore not all angles would be equal.

d False; any exponential function where time is the independent variable must have a lower limit of 0.

e True

f False; $a = -1$ and $b = 1$ also make the equation true.

3 **a** Yes, the statement is false.

b No, the statement is about blondes so it is not a counter-example.

c No, the statement is about parallelograms so it is not a counter-example (the statement is true).

d No, the statement is about similar figures so it is not a counter-example.

e No, the brackets in the statement indicate that -1 and 6 are not to be included in the domain (the statement is true).

f No, the statement indicates that that all values for x must be positive.

4 **b** Some people have blonde hair and green (or brown) eyes.

d Triangle ABC (3 cm, 4 cm, 5 cm) is similar to triangle DEF (6 cm, 8 cm, 10 cm).

f For $x = \frac{1}{2}$, $LHS = \left(\frac{1}{2}\right)^2 = \frac{1}{4}$

$RHS = \left(\frac{1}{2}\right)^3 = \frac{1}{8}$

$\frac{1}{4} \neq \frac{1}{8}$

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